FIBREWISE LOCAL COMPLETE INTERSECTION MAPS MAKE THE H^{-1} CONNECTING MAP INJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof reduces the cohomology groups to limits over the reductions via Artin-Rees completeness and lemma-NL-is-limit, applies the algebraic transitivity lemma fibrewise, and concludes injectivity of the limit map

Editorial note. The label lemma-transitive-lci-at-end names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the exact sequence of completed naive cotangent complexes for A -> B -> C (lemma-exact-sequence-NL of the same chapter)
- Artin-Rees consequence: H^{-1} of the completed complex is I-adically complete as a finite module, and equals the limit of its reductions (More on Algebra, Lemma more-algebra-lemma-consequence-Artin-Rees; Algebra, Lemma algebra-lemma-completion-tensor)
- the identification NL^wedge_{B_n/A_n} = NL^wedge_{B/A} tensor B_n (lemma-NL-is-limit of the same chapter)
- the algebraic transitivity statement: fibrewise local complete intersection maps give injectivity of the connecting map on H^{-1} (More on Algebra, Lemma more-algebra-lemma-transitive-lci-at-end)

Source reference. Original lemma 00MA: sources/stacks-chow/tags.txt, source lines 816–816. Prerequisite pointer invoked at proof line(s) 804; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0AJS: sources/stacks-restricted/pinned-original-source.tex, source lines 596–596. Prerequisite pointer invoked at proof line(s) 806; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 00S2: sources/stacks-restricted/pinned-original-source.tex, source lines 725–725. Prerequisite pointer invoked at proof line(s) 781; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 0EGU: sources/stacks-chow/tags.txt, source lines 18109–18109. Prerequisite pointer invoked at proof line(s) 801; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07D4: sources/stacks-chow/tags.txt, source lines 9399–9399. Prerequisite pointer invoked at proof line(s) 809; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *With assumptions as in Lemma 00S2 assume that $B/I^nB \to C/I^nC$ is a local complete intersection homomorphism for all n. Then $H^{-1}(NL^\wedge_{B/A} \otimes_B C) \to H^{-1}(NL^\wedge_{C/A})$ is injective.*

Proof. For each $n \geq 1$ we set $A_n = A/I^n$, $B_n = B/I^nB$, and $C_n = C/I^nC$. We have

$$\begin{aligned} H^{-1}(NL^\wedge_{B/A} \otimes_B C) &= \lim H^{-1}(NL^\wedge_{B/A} \otimes_B C_n) \\ &= \lim H^{-1}(NL^\wedge_{B/A} \otimes_B B_n \otimes_{B_n} C_n) \\ &= \lim H^{-1}(NL_{B_n/A_n} \otimes_{B_n} C_n) \end{aligned}$$

The first equality follows from More on Algebra, Lemma 0EGU and the fact that $H^{-1}(NL^\wedge_{B/A} \otimes_B C)$ is a finite C-module and hence I-adically complete for example by Algebra, Lemma 00MA. The second equality is trivial. The third holds by Lemma 0AJS. The maps $H^{-1}(NL_{B_n/A_n} \otimes_{B_n} C_n) \to H^{-1}(NL_{C_n/A_n})$ are injective by More on Algebra, Lemma 07D4. The proof is finished because we also have $H^{-1}(NL^\wedge_{C/A}) = \lim H^{-1}(NL_{C_n/A_n})$ similarly to the above. □